

ASSIGNMENT-2

AIM :- Develop a web application using JS to implement sessions, cookies, DOM. Perform validations such as checking for emptiness, only number for phone number, special character requirement for password, regular expressions for certain format of the fields etc. Use the MySQL database.

OBJECTIVES :-

- 1) To understand what form validation is.
- 2) To learn basic functioning of DOM objects.
- 3) To learn how to apply various tech. to implement it.

THEORY :-

1) **RESEX :-** Regular expressions are patterns used to match strings of text. They help validate formats like phone no. or pwd by checking if the input matches a specified pattern.

① **Why. Suitable :-** Regex provides a concise and flexible method to ensure inputs conform to specific rules. For a ph. no., regex can check if it follows a certain structure.

2) **SESSION vs. COOKIE :-**

Session :- Stores data on the server side & is temporary. It expires once the user logs out or after a set time.

Cookie :- Stores small data on the client side (user's browser), can be persistent.

3) Client-Side vs. Server-Side Validation :-

Purpose :- Client-side validation checks input before it is sent to the server, improving user experience. Server-side validation ensures that data is correct, even if the client-side validation is bypassed.

Scenario for Vulnerability :- Relying only on client-side validation is risky because a malicious user can bypass it by manipulating the browser. A hacker can send data directly to server, without client-server validation.

4) JS Interacting with DOM :-

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h2>Form Submission </h2>
```

```
<form id = "myForm">
```

```
<input type = "text" id = "name" name = "name">
```

```
<input type = "Submit" value = "Submit">
```

```
</form>
```

```
<p id = "message"></p>
```

```
<script>
```

```
document.getElementById('myForm').onsubmit = function(event)
```

```
{ event.preventDefault();
```

```
var name = document.getElementById('name').value;
```

```
document.getElementById('message').innerText = "Hello, " +  
name + "!";
```

```
};
```

```
</script>
```

```
</body>
```

```
</html>
```


Explanation:- The script changes the content of the paragraph dynamically after the form is submitted, displaying a greeting with the user's name.

5) FRONTEND TO MySQL CONNECTIVITY :-

- 1) Frontend :- HTML + CSS + JS for user input & interaction.
- 2) AJAX Request :- Use JS (via AJAX) to send data to server-side.
- 3) PHP Backend :
 - Receive data from the Frontend.
 - Connect to MySQL using mysqli connect or PDO.
 - Query the database.
- 4) MySQL :- Query execution on the database.
- 5) Return Data :- PHP returns the result back to the frontend.
- 6) Update Frontend :- Use JS to update the page based on the server's response.

FAQ'S :-

Q1) Write 3 reasons why Form validations are important.

Ans.  Data Integrity :- Ensures that the data entered by users is in the correct format (Eg. email, phone number).

User Experience :- Provides immediate feedback to users, helping them correct mistakes before submitting the form.

Security :- Prevents malicious inputs (Eg. SQL injection, XSS) by validating & sanitizing the data before processing.

Q2) Give an example of how to modify an attribute value using DOM.

Ans. `<!DOCTYPE html>`

`<html>`

`<body>`

``

`<script>`

`document.getElementById('myImage').setAttribute('src', 'image2.jpg');`

`</script>`

`</body>`

`</html>`

Explanation: This Java Script code changes the "src" attribute of the image from image1.jpg to image2.jpg.

Q3) What are the different features of Java Script?

Ans. Dynamic Typing :- Variables can hold different data type during execution.

Event Handling :- Allows interaction with HTML elements.

BOM Manipulation :- Allows dynamic interaction with HTML & CSS.

OOP :- Supports classes, inheritance, & objects.

BA
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